DATA3 · NF270 SINGLE-SALT CAMPAIGN

Identifiability & Analysis — Follow-up

Paper-justified measurement error · σ-identifiability sensitivity · axis convention · B(ionic strength) & per-band θ

Keshav Kasturi Rangan

Post-meeting analysis · June 11, 2026

The cF weight was physically unjustified — and σ depends on it

Paper: Lilonfe et al., 'Soft Sensors...', ChemRxiv 2026 (doi:10.26434/chemrxiv.15002541/v1)

Per-channel measurement error (DATA3)

Retentate cF — **0.3% → 2%**

conductivity soft-sensor MAPE ⊕ cell-const

Permeate cV — **3% (kept)**

ICP-OES reference — already physical

Mass m — 0.01 g (kept)

balance resolution

σ flips with the weight (multistart-verified)

MC3 SNaCl: σ 0.42 (interior, 0.3%) → 1.0 (wall, 2%)

MC5 S2NaCl: σ 0.0 (wall, 0.3%) → 0.35 (interior, 2%)

σ's global minimizer is weight-dependent → σ is poorly identified.

Takeaway

The interior σ we reported was partly an **artifact of the over-tight 0.3% cF weight**. With the honest 2% error, σ is revealed to be **not robustly identifiable** from single-salt runs.

→ This is a sensitivity result, not a fit fix. σ needs better EXPERIMENTS (high-cF, multi-ΔP DoE), not reweighting. The weight now feeds objective + Fisher-information + parmest consistently; DATA1/DATA2 byte-identical.

Where this points: B(ionic strength) and per-band θ

Concentration-dependent transport, made salt-transferable and band-resolved

$\beta(c) \to \beta(I)$

- $B = J\_w[\beta_0 + \beta_1 \cdot c + \beta_2 \cdot c^2]$
- plot β_0/β_1 vs L_p or σ (not lumped B)
- $I = k\_I \cdot c$ (NaCl 1 · $CaCl_2$ 3 · $LaCl_3$ 6)
- B(I) ≡ B(c) per salt (β rescaled), but β become transferable across salts → multisalt

Per-band / per-vial θ

- group vials by cF band → one θ per band
- mask the objective to a band's vials
- `B_form='pervial'` already gives per-vial B
- correlate band θ with contour basins → σ-drift with concentration

Axis convention

- Y-priority $L_p > B > \sigma$
- animations + contours regenerated
- σ-scrub now $L_p \times B$ (L_p on Y)
- rule of thumb going forward

THE GOAL

Different experiments pin different parameters — use that

Each run identifies some of $\theta = (L_p, B, \sigma)$ well and others not at all. Two questions follow:

Q1 · Combine

Can we combine experiments into a better **initial guess** for the joint fit — taking each parameter from the run that actually constrains it?

Q2 · Design

Which **new experiments** would break the correlations that leave σ and B unidentified for the harder salts?

The logic flow of this talk

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Model	**Diagnose**	**Read where θ is pinned**	**Combine → seed**	**Design new experiments**
$\theta = (L_p, B, \sigma)$ in Spiegler–Kedem	WSSE contour maps + animations	good / okay / poor per experiment	better initial guess per salt	break σ–B, pin L_p

Takeaway up front: the campaign already contains everything we need to seed every salt's fit — and tells us exactly which two experiments would close the gaps.

What we simulate and what we estimate

Spiegler–Kedem transport through the NF membrane — three parameters per experiment: $\theta = (L_p, B, \sigma)$

Water flux · Eq. (1)

σ scales how much osmotic back-pressure opposes the applied ΔP

$$Jw = Lp\,(\Delta P - \sigma\,\Delta\pi)$$

Osmotic pressure · van 't Hoff

n = dissolved species per formula unit (NaCl 2 · $CaCl_2$ 3 · $LaCl_3$ 4)

$$\Delta\pi = n\,R\,T\,(c\_in - c\_h)$$

Concentration polarization · Eq. (3)

wall enrichment from the boundary layer (mass-transfer coefficient k)

$$c\_in = (c\_f - c\_h) \cdot \exp(Jw / k) + c\_h$$

Solute flux · Eq. (2)

diffusion-dominant; B = solute permeability coefficient

$$Js = B\,(c\_in - c\_h)$$

Same notation as the DATA1 / DATA2 papers. Concentration fits curve because the state ODEs are nonlinear; DATA3 also allows B = B(c_f) **`(B_form = 1)`** when a constant B underfits.

01 · CONFIRM FIRST

The parameter bounds I'm changing

L_p and σ inherited from the DATA1/DATA2 KCl baseline — the per-salt B upper bound is the only new knob

Inherited baseline (KCl · DATA1/DATA2)

L_p ∈ [0.5, 50] L/(m²·hr·bar)

σ ∈ [0, 1] (dimensionless reflection coeff.)

B ∈ [0, 30] µm/s — unchanged for monovalent

New: per-salt B upper bound — halve per +1 cation charge

NaCl	[0, 30] µm/s	monovalent — same as KCl
$CaCl_2$	[0, 15] µm/s	divalent — halved
$LaCl_3$	[0, 10] µm/s	trivalent — a third

How the contour panels are built

At every (σ, L_p) grid point we forward-simulate the full Spiegler–Kedem DAE (B fixed at warm-start, S_0/S at warm-start) then evaluate the weighted sum-of-squared-residuals for each channel.

5 channels per panel: mass · permeate conc. · retentate conc. · permeate conductivity · retentate conductivity

This is a diagnostic, not a fit — pure forward-simulation. Where the 5 channels agree on a minimum, the data identify the parameters; where they split, they don't.

02 · DIAGNOSE

Map every parameter, then read what's identified

Forward-simulated WSSE contours + animations — good · okay · poor, per experiment

Contour maps = forward-simulated WSSE, no fitting

We sweep a parameter grid, simulate the full DAE at each node, and score the weighted residual per channel

Weighted sum of squared residuals, summed over the 3 fitted channels (mass · permeate c_v · retentate c_r)

$$\mathrm{WSSE}(\theta) = \Sigma\ \mathrm{ch}\in\{m,cv,cr\}\ \ (1/N\_ch)\ \Sigma\_i\ (\ y\_i\ \mathrm{model} - y\_i\ \mathrm{exp}\ )^2$$

Best fit

$$\theta^* = \mathrm{argmin}\_\theta\ \ \mathrm{WSSE}(\theta)$$

$\theta = (L_p, B, \sigma)$. If the minimum is a sharp interior point, θ is identified; if it lies on a wall or in a flat valley, it is not.

Contour-consistent seed

$$\theta\_init = \mathrm{argmin}\ (\sigma,Lp)\ [\ \mathrm{WSSE\_m} + \mathrm{WSSE\_cv} + \mathrm{WSSE\_cr}\ ]$$

Read the joint 2-D minimum straight off the contour (B, S_0, S held at warm-start) and use it to seed the full fit.

Two views of the same objective, both shown in this deck

- **Static $\sigma \times L_p$ panels** — all 5 measurement channels at once (mass, perm/ret conc, perm/ret conductivity). Where channels agree → trust.
- **Animated scrubs** — hold one parameter and sweep it; watch whether the 2-D basin for the other two stays put. This is how to read a good initial guess for each experiment.

  ① $B \times L_p$ basin as $\sigma : 0 \to 1$ ② $\sigma \times L_p$ basin as $B : 0 \to B\_max$

Three stages of identifiability across all 11 sheets

Agreement across the 5 measurement channels = trust. What we trust most → what we trust least.

STAGE 1

Well-identified

2 sheets · NaCl

Both are diluting NaCl runs that start at high cf (~95 mM). High concentration → large Δπ → σ is elucidated (DATA1 §4.3.2), and L_p is pinned by mass. MC3 SNaCl keeps an interior σ; MC5 S2NaCl's σ drifts but L_p holds.

fit quality: GOOD

STAGE 2

Inconsistent

7 sheets · 4 NaCl + 3 $CaCl_2$

L_p tight, but σ rail-pinned at a wall. Channels each find a minimum, but at different (σ, L_p) walls — a weighting split, not a model-form failure.

fit quality: OKAY

STAGE 3

Not estimable

2 sheets · $LaCl_3$

Same wrong-model bias on both sheets; WSSE ~20× the NaCl gold standard. Constant-(σ,B) Spiegler–Kedem can't represent La^{3+} ion-pairing.

fit quality: POOR

Where the optimizer landed on every sheet

Warm-start θ across the campaign, colour-coded by stage. ↑/↓ = parameter pinned at a bound.

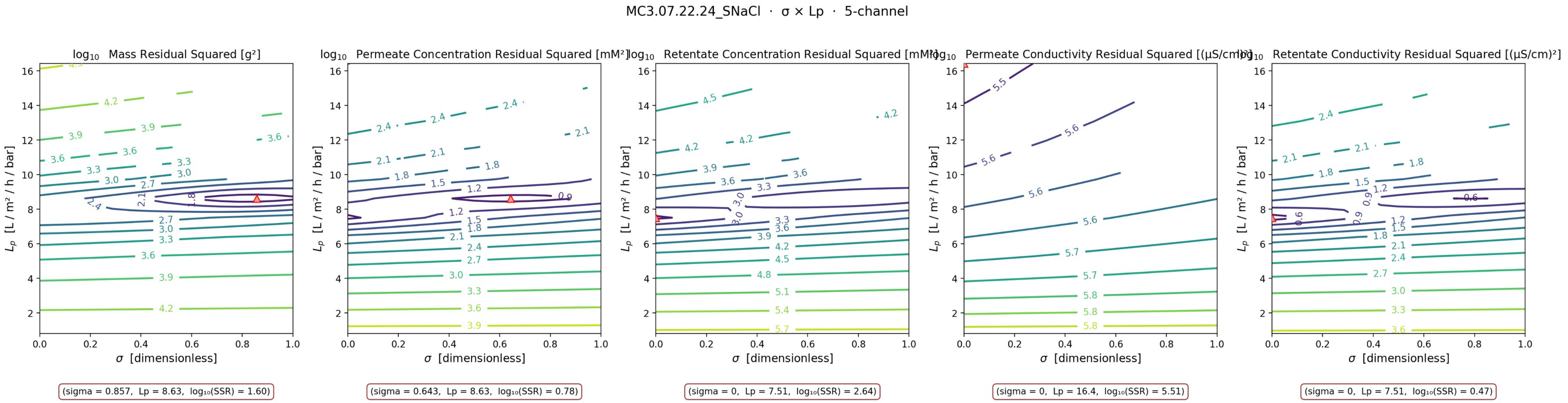
Sheet	L_p	B [µm/s]	σ	$WSSE_3$ch	Stage assignment
MC3 · 07.22.24 · SNaCl	8.63	14.68	0.79	560	**Stage 1A · interior σ — gold standard**
MC5 · 07.23.24 · S2NaCl	7.75	10.14	0.29	888	**Stage 1B · diluting, σ harmless**
MC4 · 07.11.24 · SNaCl	11.35	2.02	0.50	19,508	**Stage 1B-ish · diluting BUT poor WSSE ⚠**
MC2 · 05.07.24 · NaCl	10.45	9.73	**1.00 ↑**	4,528	**Stage 2 · σ wall, concentrating**
MC5 · 07.23.24 · NaCl	10.52	3.70	0.79	7,024	**Stage 2 · σ wall, concentrating**
MC5 · 07.23.24 · SNaCl	8.64	3.07	0.57	5,870	**Stage 2 · σ wall, concentrating**
MC2 · 05.07.24 · $CaCl_2$	8.00	0.51	0.50	97,213	**Stage 2 · channels split by side**
MC3 · 07.11.24 · $SCaCl_2$	6.56	9.11	**1.00 ↑**	6,139	**Stage 2 · σ at wall**
MC3 · 07.12.24 · $S2CaCl_2$	5.49	30.00 ↑	**1.00 ↑**	9,774	**Stage 2 · σ AND B at walls**
MC2 · 05.21.24 · $LaCl_3$	2.95	0.31	**0.00 ↓**	21,500	**Stage 3 · wrong model — all channels disagree**
MC4 · 07.11.24 · $SLaCl_3$	5.55	0.17	0.93	75,080	**Stage 3 · wrong model — replicate**

Only MC3 SNaCl landed at an interior σ — that is the reference 'well-identified' case.

GOOD — interior σ from a high-cf diluting run (gold standard)

Identifiability: GOOD

MC3 · 07.22.24 · SNaCl · σ × L_p, 5 channels



Legend: ▲ = per-panel minimum (argmin) **contour lines** = $\log_{10}$ weighted SSR (cooler → better fit) each panel = one measurement channel panels agree → identifiable

θ here: Lp = 8.63, B = 14.68 µm/s, σ = 0.79, WSSE_3ch = 560

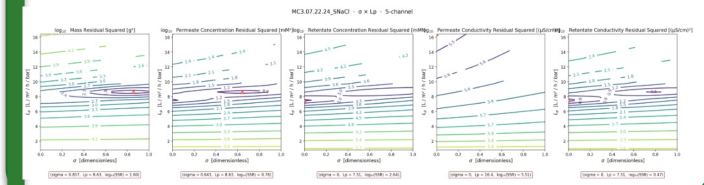
A diluting NaCl run starting at cf ≈ 95 mM, so Δπ is large early and σ leaves a real fingerprint (DATA1 §4.3.2). All 5 channels at the warm-start B; mass and permeate-concentration both argmin at an interior σ near L_p ≈ 8.6. This is the only NaCl sheet where σ is genuinely identified, and the cleanest fit in the campaign.

Every sheet's σ × L_p contour, colour-tabbed by stage

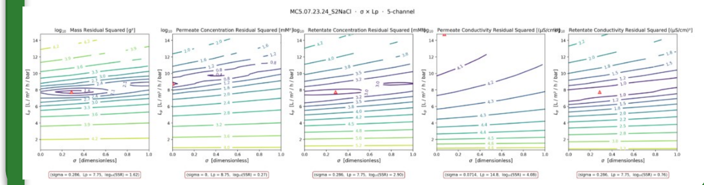
Basin consistent (Stage 1) → split (Stage 2) → absent (Stage 3). Each tile: 5 channels, contour = $\log_{10}$ WSSE, ▲ = argmin.

Stage 1 well-id. | Stage 2 inconsist. | Stage 3 not-est.

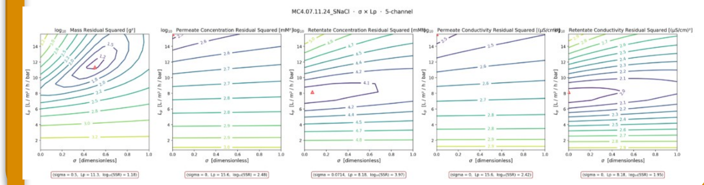
MC3 · 07.22.24 · SNaCl



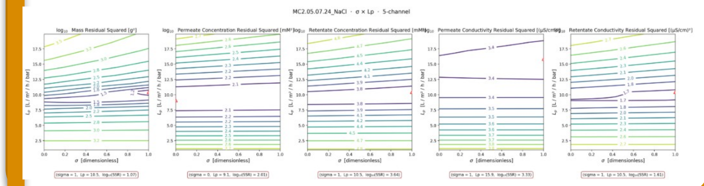
MC5 · 07.23.24 · S2NaCl



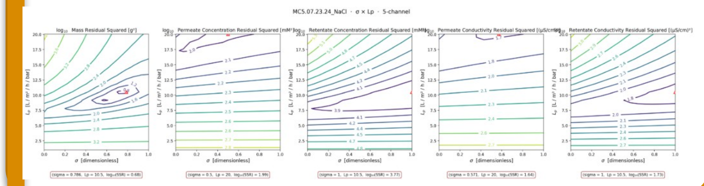
MC4 · 07.11.24 · SNaCl



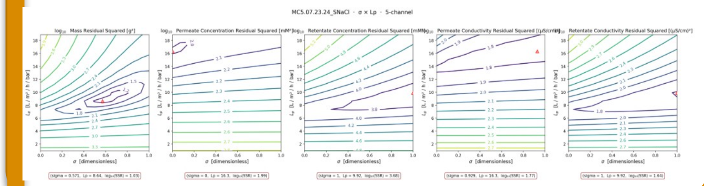
MC2 · 05.07.24 · NaCl



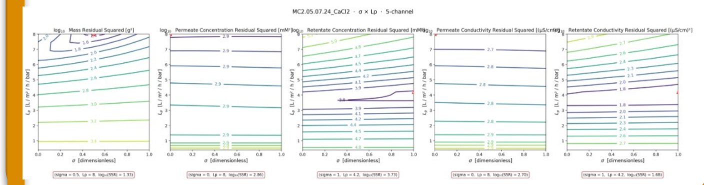
MC5 · 07.23.24 · NaCl



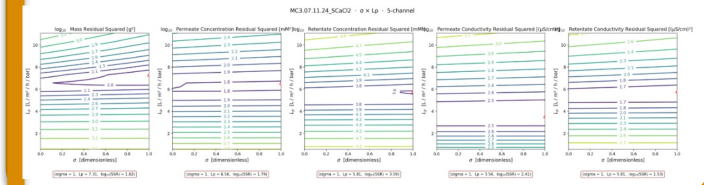
MC5 · 07.23.24 · SNaCl



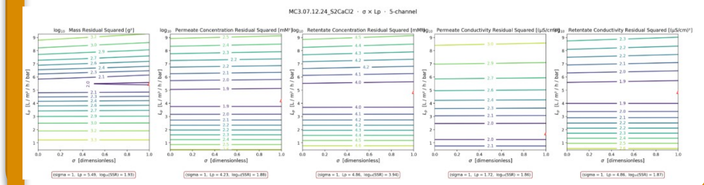
MC2 · 05.07.24 · $CaCl_2$



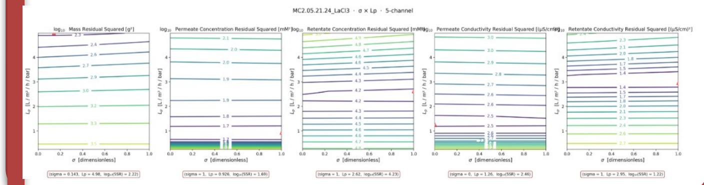
MC3 · 07.11.24 · $SCaCl_2$



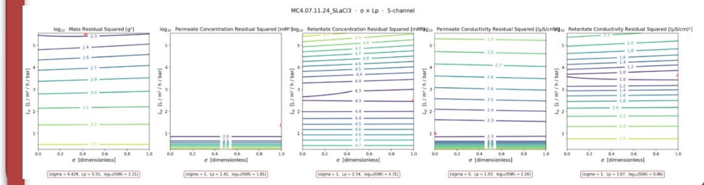
MC3 · 07.12.24 · $S2CaCl_2$



MC2 · 05.21.24 · $LaCl_3$



MC4 · 07.11.24 · $SLaCl_3$



σ leaves a fingerprint only when osmotic pressure is large

The sensitivity of the data to σ is set by Δπ — so the operating regime decides what each run can identify

σ enters the model only through σ·Δπ, so

$$\partial Jw/\partial\sigma = -Lp\,\Delta\pi = -Lp \cdot n\,R\,T\,(c\_in - c\_h)$$

High c_f → σ visible

$$\Delta\pi = n\,R\,T\,cf \quad (c\_h \approx 0)$$

Large cf → large Δπ → ∂Jw/∂σ large → σ leaves a fingerprint (DATA1 §4.3.2). Both gold sheets are diluting runs starting at cf ≈ 95 mM.

Low c_f → L_p only

$$cf \to 0 \;\; \Delta \qquad PJw = Lp$$

Δπ → 0: L_p reads directly from mass, independent of σ and B. This is a pure-water run, or the end of any diluting run.

σ–B coupling

$$dc\_f/dt = (A\_m \,/\, m\_f)(Jw\,c\_d - Js)$$

σ acts only through Jw, B only through Js — but both drive c_f(t). Vary ΔP (hence Jw) and their relative pull changes, separating them.

So: a diluting run from high cf pins L_p AND σ; a concentrating run from low cf pins only L_p; a pressure sweep separates σ from B.

03 • IN MOTION

Read the initial guess off the moving basin

Two sweeps per experiment: ① $B \times L_p$ as $\sigma{:}0 \to 1$ ② $\sigma \times L_p$ as $B{:}0 \to B_{max}$

Watch the objective basin move

Same 5-channel diagnostic, animated — hold one parameter fixed and sweep it frame by frame

How to read the scrub animations

Each frame fixes one parameter and shows the objective basin for the other two, across 3 channels (mass · permeate-conc · retentate-conc). The red ▲ marks the per-frame minimum.

Read the motion as identifiability: a basin that stays put and the 3 channels agree → **identifiable**. A minimum that slides along a wall, or channels that disagree → **not**.

GOOD — basin stays, channels agree	OKAY — minimum slides to a wall	POOR — no coherent basin

▶ *These slides contain animated GIFs — they play automatically in PowerPoint slideshow mode. The σ-sweep ($B \times L_p$ basin moving through σ) is the one to watch: it is the animated twin of the static $\sigma \times L_p$ panels.*

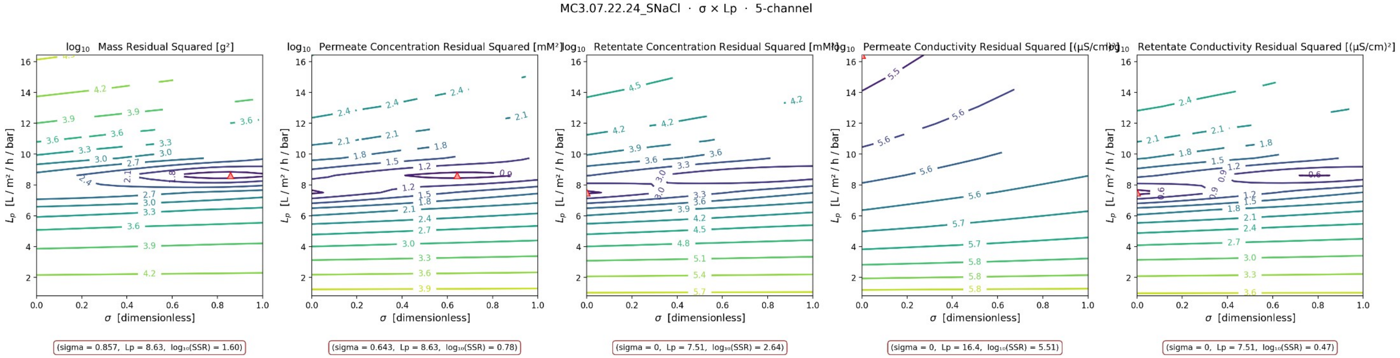
STAGE 1 · WELL-IDENTIFIED

Identifiability: GOOD

GOOD — the basin holds under both sweeps

MC3 · 07.22.24 · SNaCl · read the joint minimum off both sweeps

① L_p × B basin as σ : 0 → 1 (L_p on Y)



② σ × L_p basin as B : 0 → B_max

(σ × L_p sweep renders once the 3-D grid finishes)

Initial guess to read off

$Lp \approx 8.6$

$B \approx 14.7\ \mu m/s$

$\sigma \approx 0.79$

joint argmin of mass + c_v + c_r

Across both sweeps the minimum stays anchored (L_p ≈ 8.6, interior σ) and the channels track together. A sharp, stationary basin = a trustworthy seed. This is the gold-standard read: take all three parameters from this run.

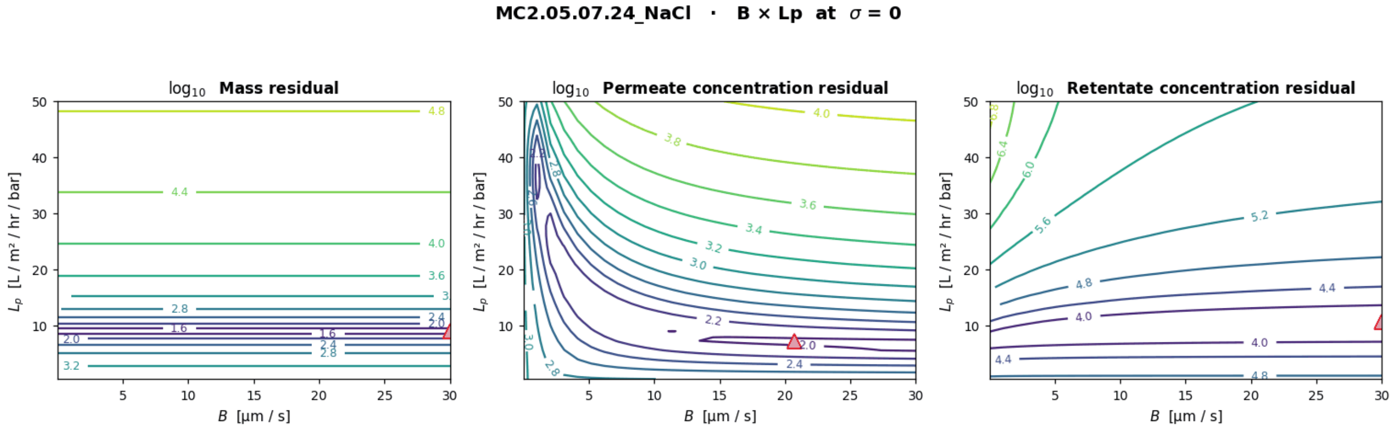
Legend: ▲ = per-panel minimum (argmin) **contour lines** = $\log_{10}$ weighted SSR (cooler → better fit) each panel = one measurement channel panels agree → identifiable

Identifiability: OKAY

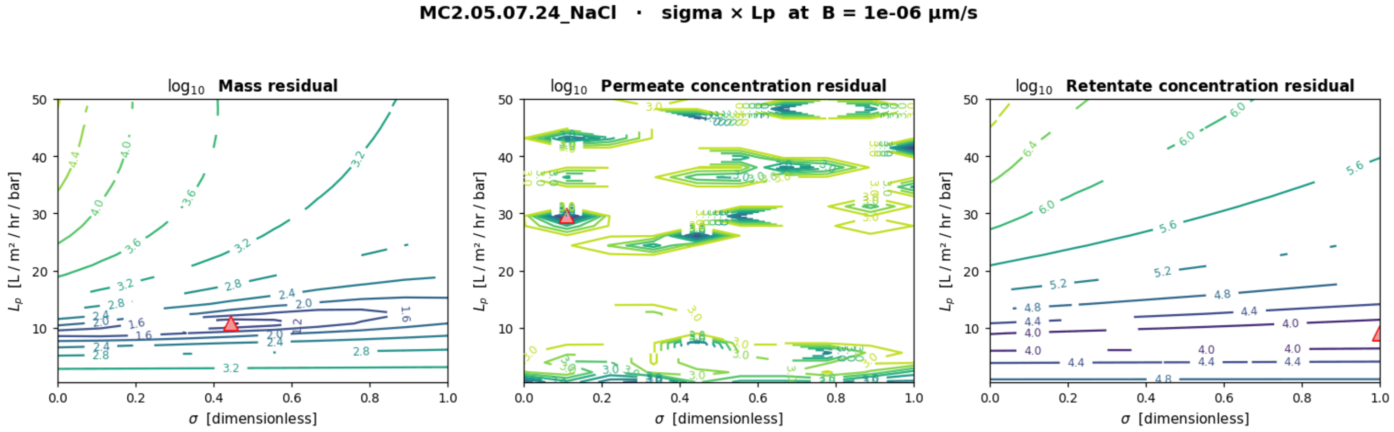
OKAY — L_p holds, σ slides to the wall

MC2 · 05.07.24 · NaCl · read the joint minimum off both sweeps

① L_p × B basin as σ : 0 → 1 (L_p on Y)



② σ × L_p basin as B : 0 → B_max



Legend: ▲ = per-panel minimum (argmin) **contour lines** = $\log_{10}$ weighted SSR (cooler → better fit) each panel = one measurement channel panels agree → identifiable

Initial guess to read off

Lp ≈ 10.5

B ≈ 9.7 µm/s

σ ≈ 1.00

joint argmin of mass + c_v + c_r

Sweep ① : the B×L_p minimum keeps the same L_p but drifts to the σ=1 wall. Sweep ② : as B varies the σ×L_p minimum barely moves in L_p — L_p is solid. Take L_p from here; do NOT trust σ — get σ from the gold-standard run instead.

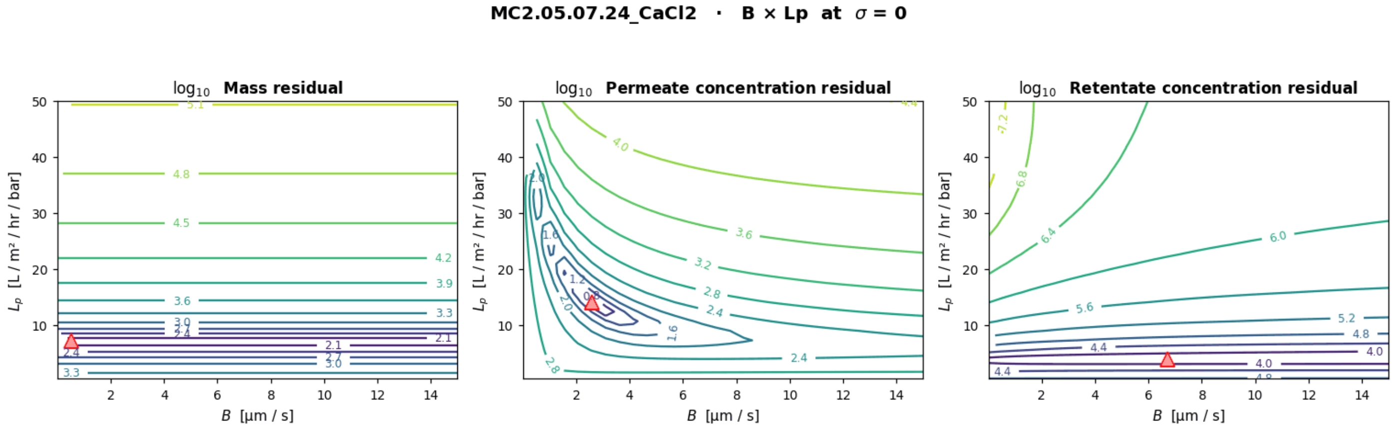
STAGE 2 · INCONSISTENT ($CaCl_2$)

Identifiability: OKAY

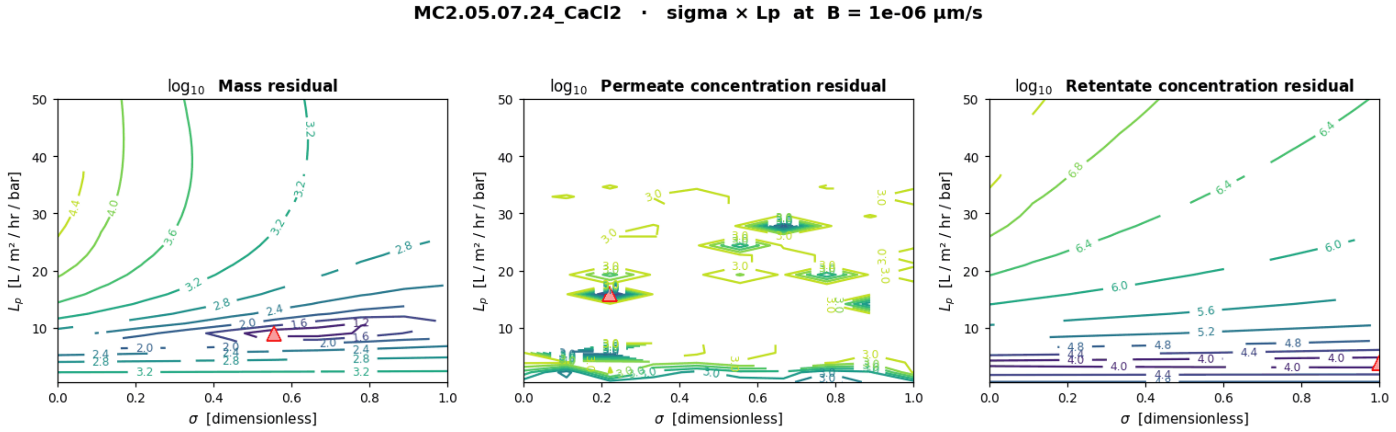
OKAY — L_p usable, σ and B trade off

MC2 · 05.07.24 · $CaCl_2$ · read the joint minimum off both sweeps

① L_p × B basin as σ : 0 → 1 (L_p on Y)



② σ × L_p basin as B : 0 → B_max



Legend: ▲ = per-panel minimum (argmin) **contour lines** = $\log_{10}$ weighted SSR (cooler → better fit) each panel = one measurement channel panels agree → identifiable

Initial guess to read off

Lp ≈ 8.0

B ≈ 0.5 µm/s

σ ≈ 0.50

joint argmin of mass + c_v + c_r

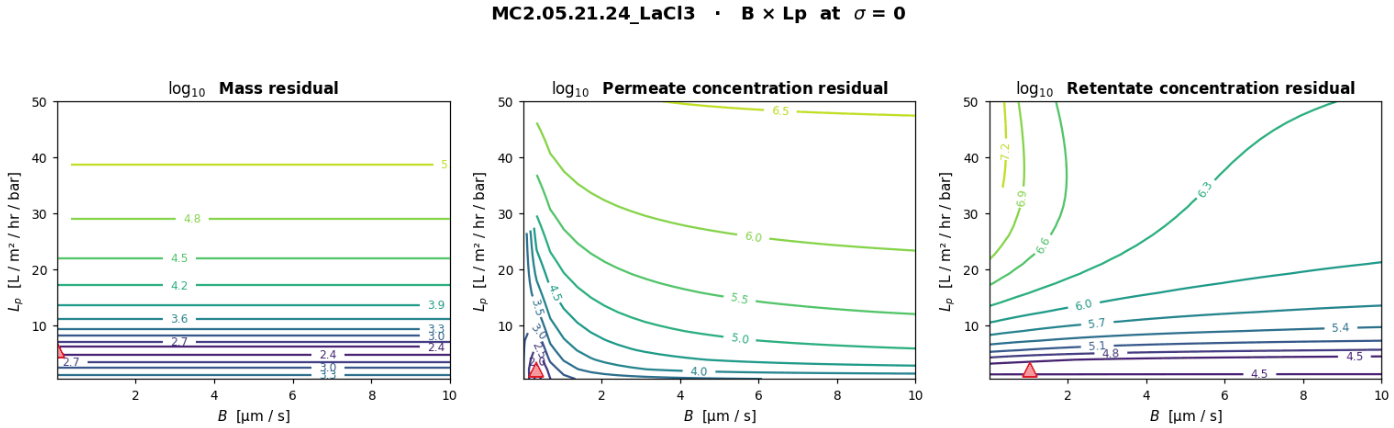
The two sweeps show the σ–B trade-off directly: moving B shifts where σ wants to sit. L_p is still the reliable read; σ and B are correlated and need a pressure sweep to separate. Seed L_p from here, σ from the σ-probing run, B from its bound.

Identifiability: POOR

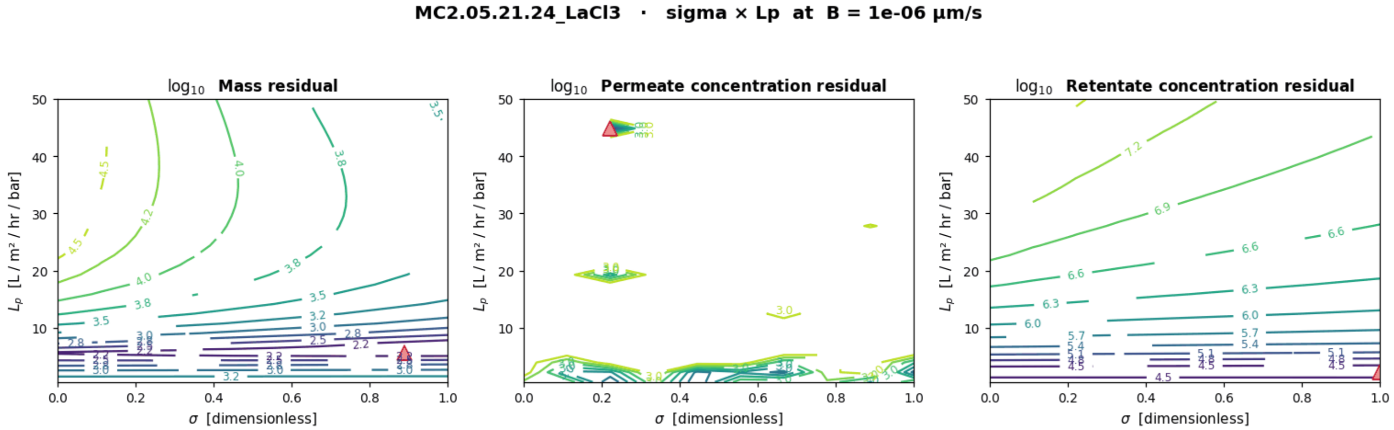
POOR — no stationary basin under either sweep

MC2 · 05.21.24 · $LaCl_3$ · read the joint minimum off both sweeps

① L_p × B basin as σ : 0 → 1 (L_p on Y)



② σ × L_p basin as B : 0 → B_max



Legend: ▲ = per-panel minimum (argmin) **contour lines** = $\log_{10}$ weighted SSR (cooler → better fit) each panel = one measurement channel panels agree → identifiable

Initial guess to read off

$Lp \approx 3.0$

$B \approx 0.3$ µm/s

$\sigma \approx 0.00$

joint argmin of mass + c_v + c_r

Neither sweep produces a consistent minimum and the channels never agree — the constant-(σ,B) model can't represent La^{3+}. Don't read a seed from this run; it needs a model fix (ion-pairing) and more replicates before any θ is meaningful.

04 • FITS

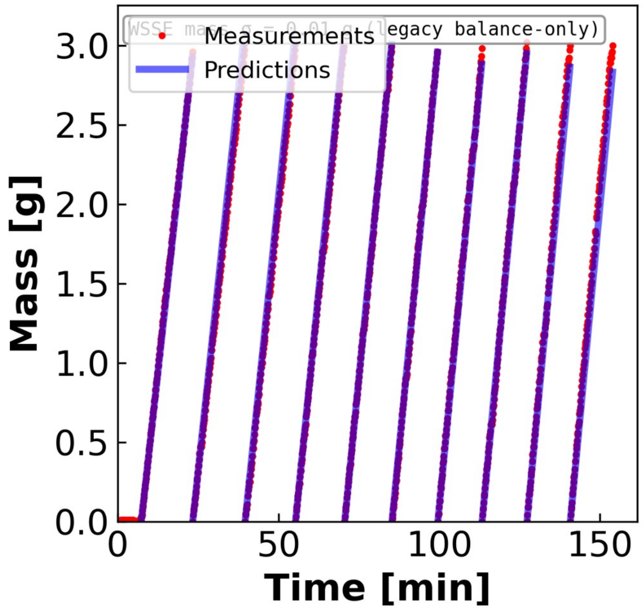
Mass & concentration fits from the contour seed

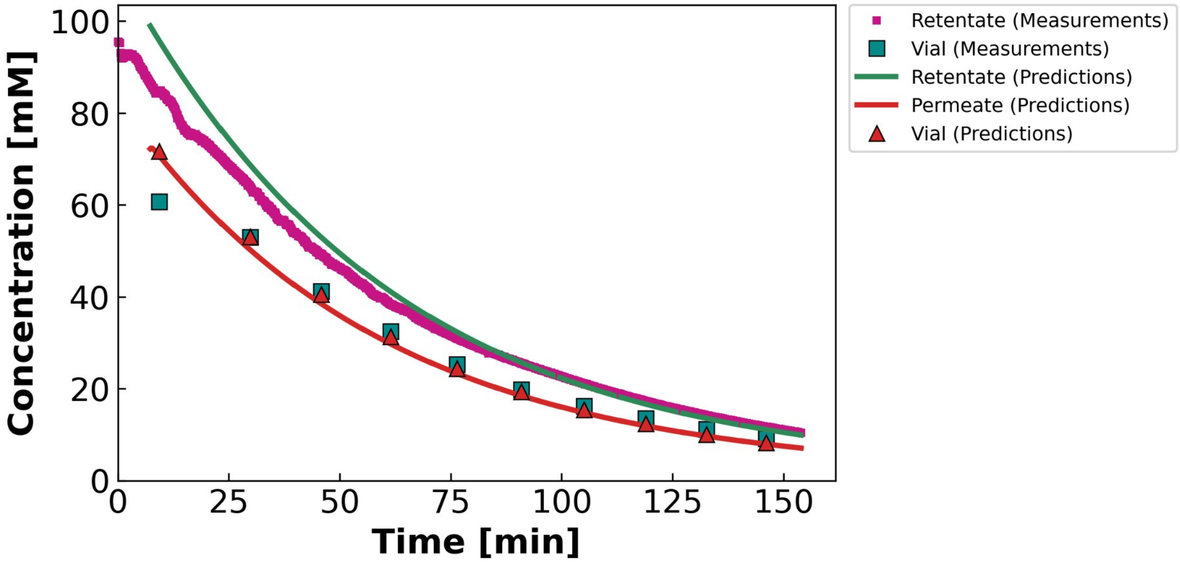
θ from the joint contour argmin · variable B (B_form = 1) → curved $c_h(t)$

MC3 · 07.22.24 · SNaCl

STAGE 1A · well-identified

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 8.63

B = 14.68 μm/s

σ = 0.79

WSSE₃ch

560

Identifiability: GOOD

Gold standard. Interior σ → both mass and c^f tracked. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

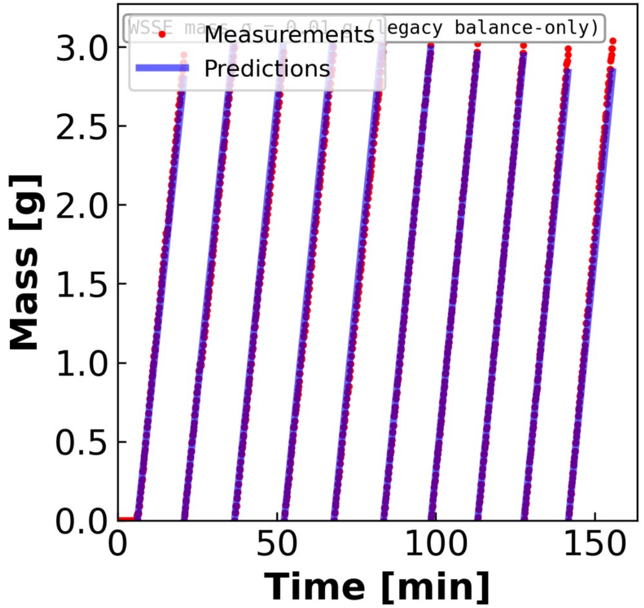
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

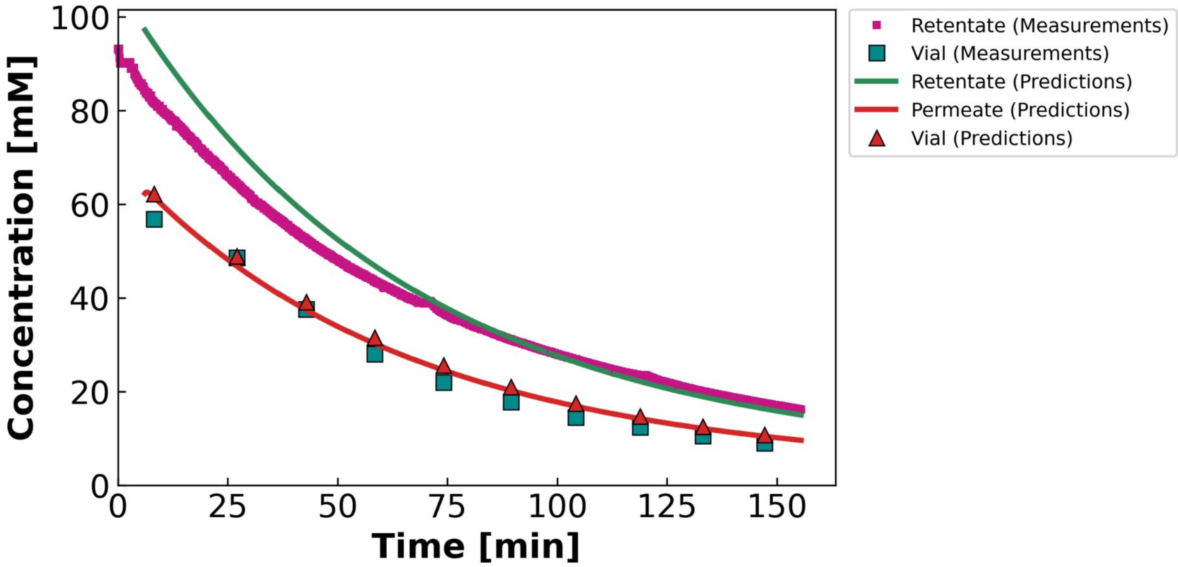
04 · CONTOUR-CONSISTENT FIT

MC5 · 07.23.24 · S2NaCl

STAGE 1B · diluting

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

L_p = 7.75

B = 10.14 µm/s

σ = 0.29

$WSSE_{3}ch$

888

Identifiability: GOOD

Diluting run → c^f falls. L_p pinned by mass; σ irrelevant. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

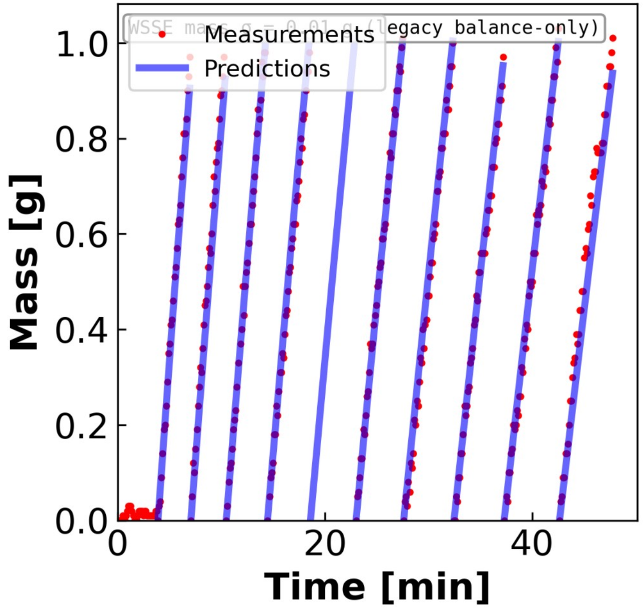
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

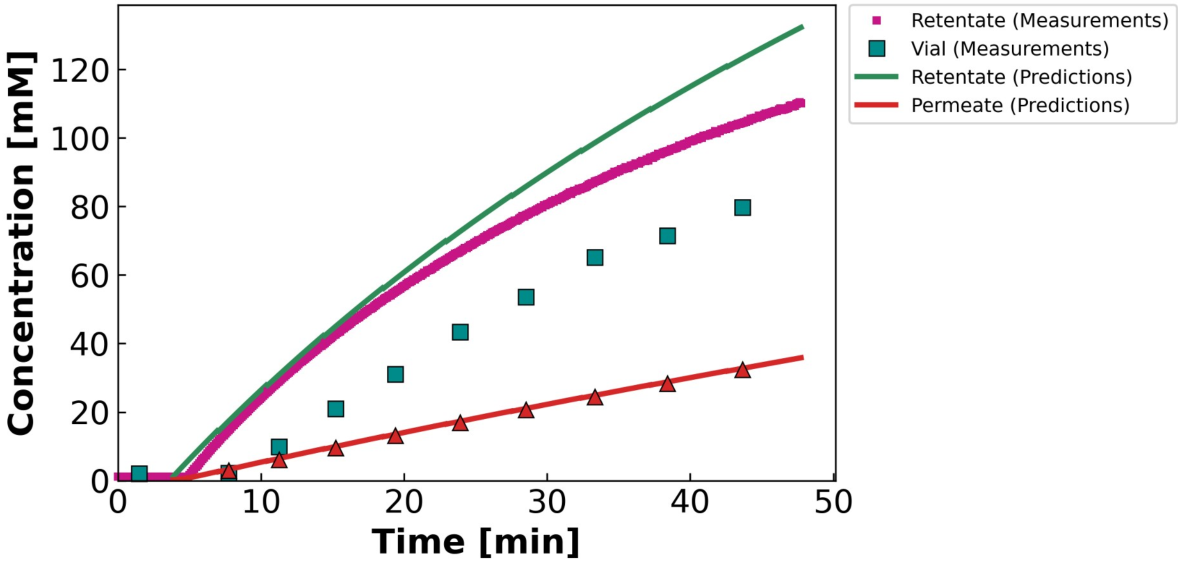
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 1B · diluting ⚠

MC4 · 07.11.24 · SNaCl

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 11.35

B = 2.02 µm/s

σ = 0.50

$WSSE_{3ch}$

19,508

Identifiability: OKAY

Diluting but WSSE ~10× worse — suspected data-quality issue. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

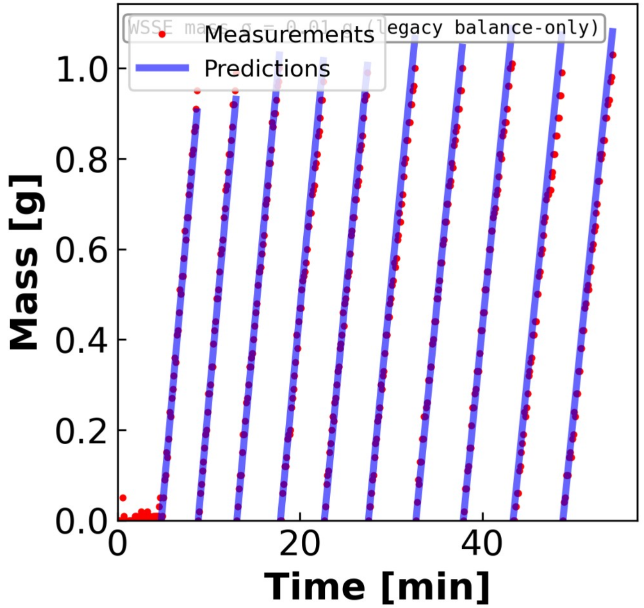
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

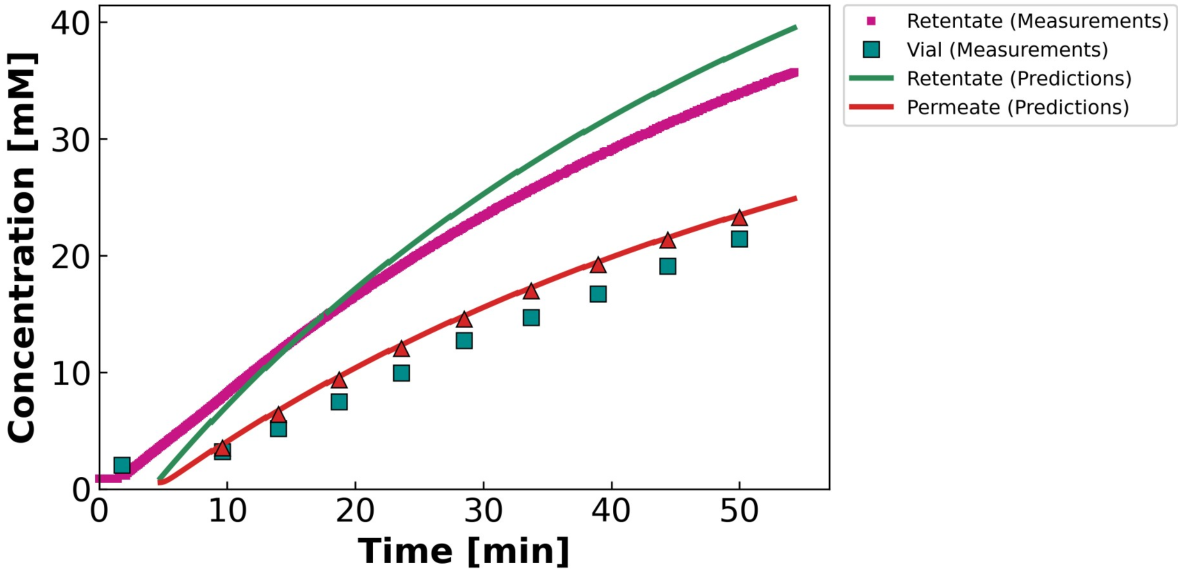
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 2 · NaCl, σ=1 wall

MC2 · 05.07.24 · NaCl

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 10.45

B = 9.73 μm/s

σ = 1.00 ↿

$WSSE_3ch$

4,528

Identifiability: OKAY

Mass fit good; concentration biased by σ pinned at 1. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

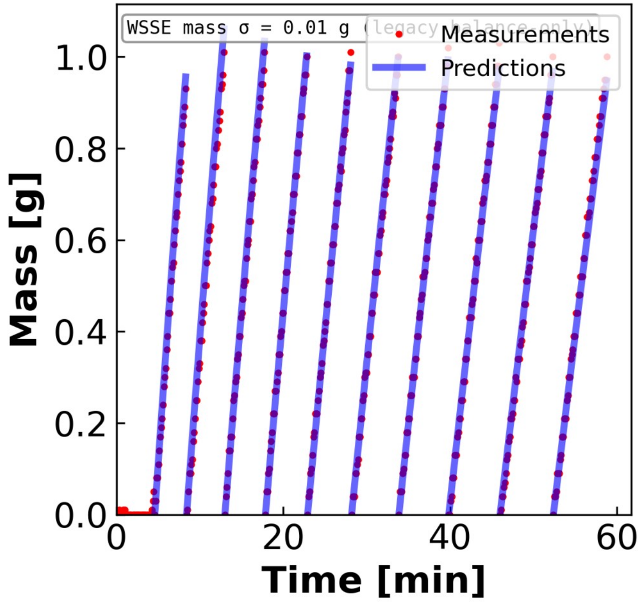
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

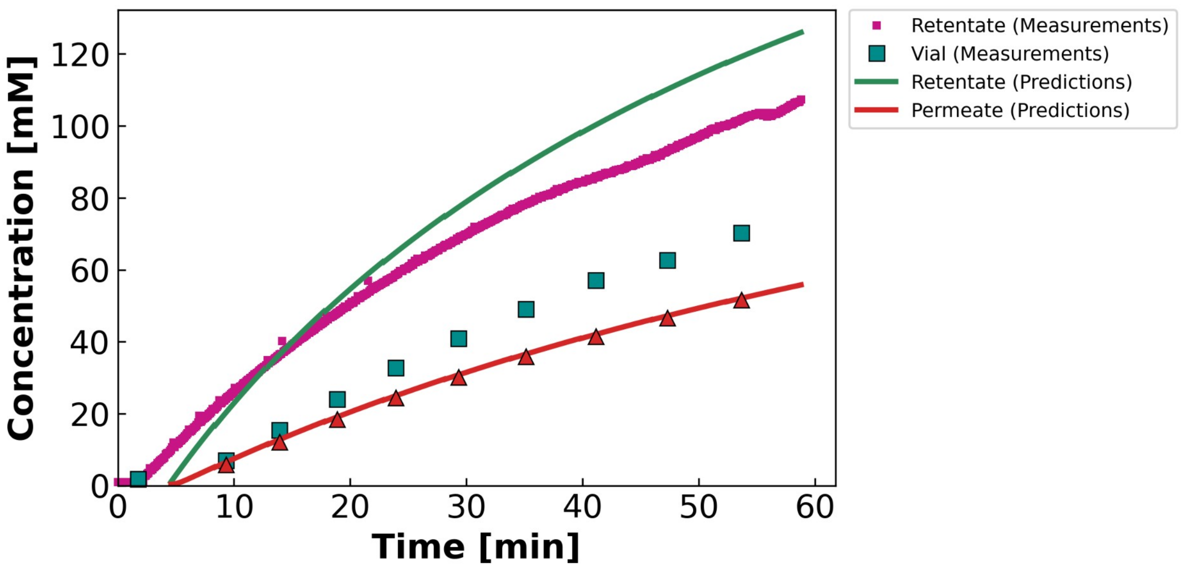
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 2 · NaCl, σ=1 wall

MC5 · 07.23.24 · NaCl

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 10.52

B = 3.70 μm/s

σ = 0.79

$WSSE_3$ch

7,024

Identifiability: OKAY

Concentrating; σ at wall. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

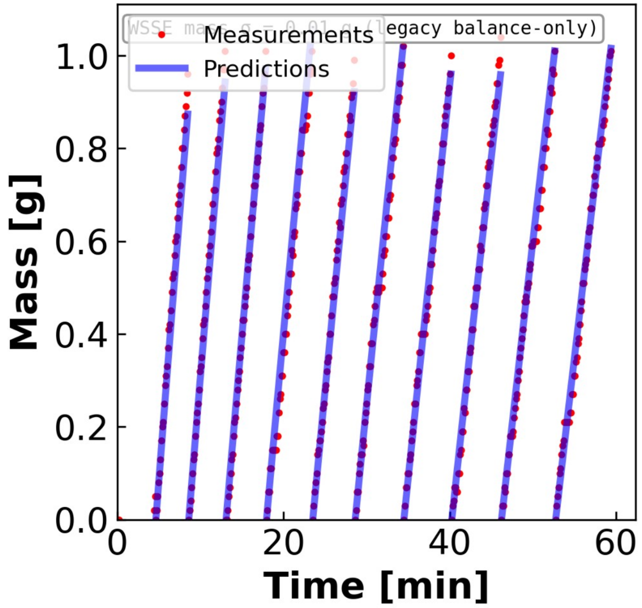
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

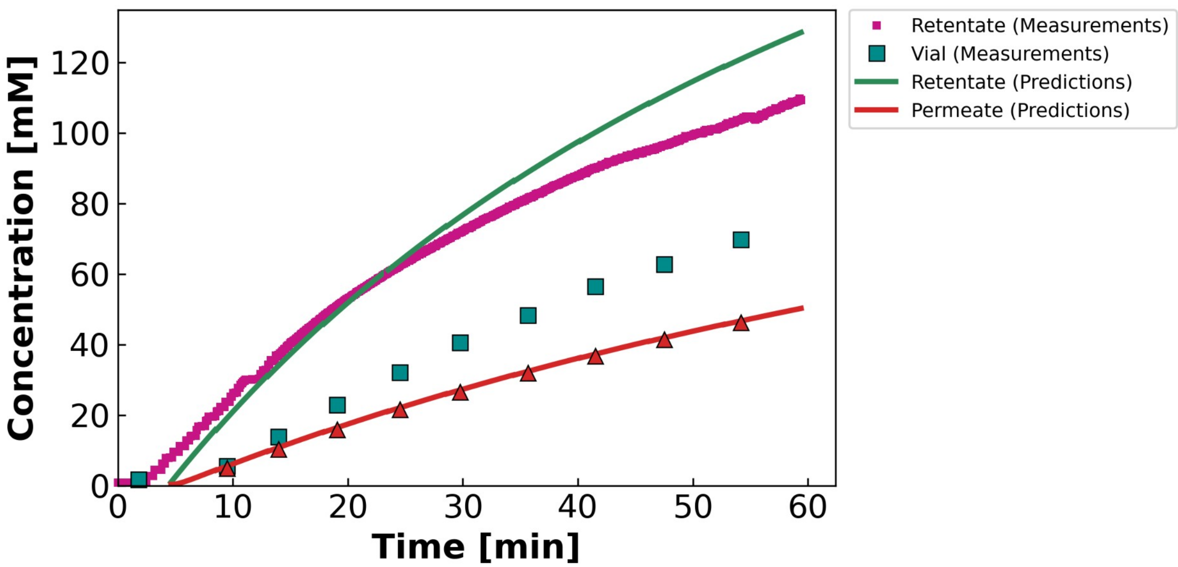
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

MC5 · 07.23.24 · SNaCl

STAGE 2 · NaCl, σ=1 wall

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 8.64

B = 3.07 μm/s

σ = 0.57

WSSE$_3$ch

5,870

Identifiability: OKAY

Concentrating; σ at wall. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

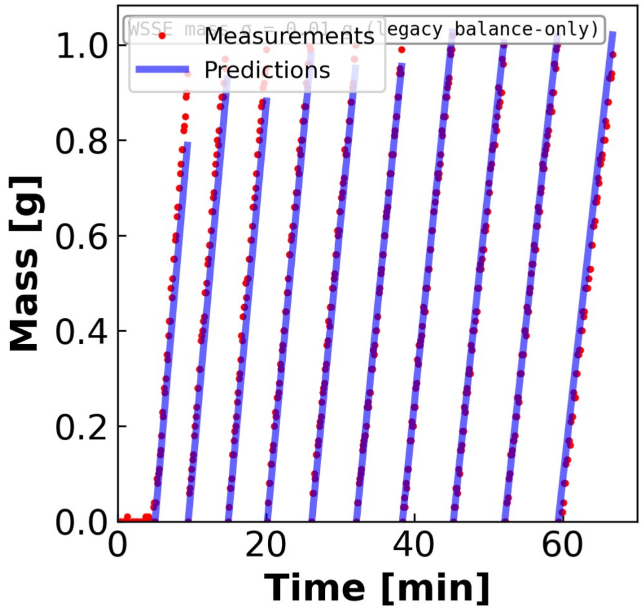
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

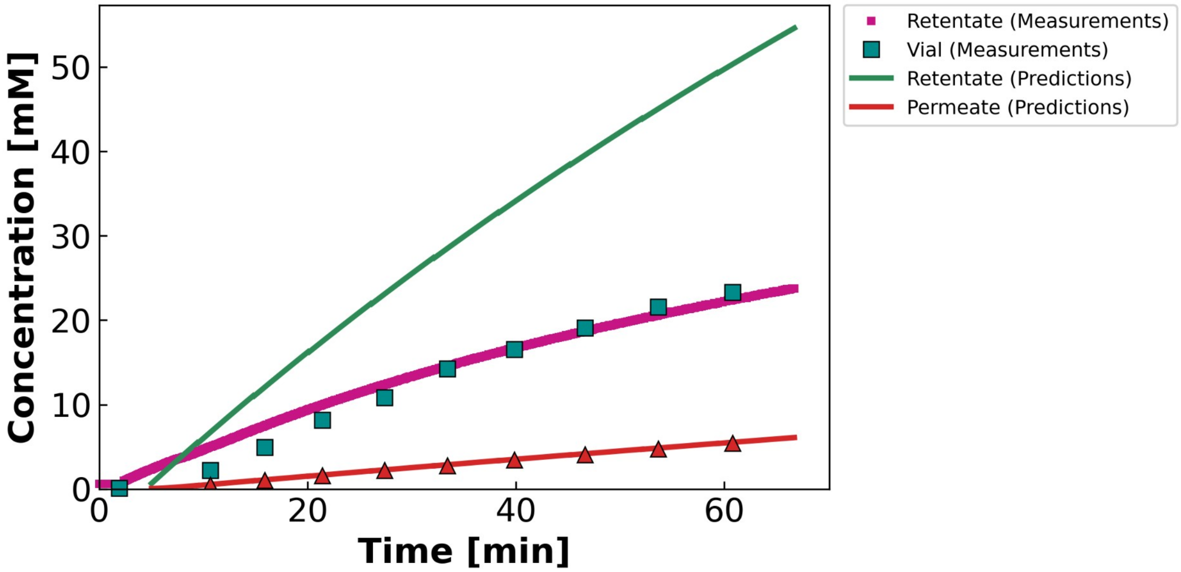
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 2 · $CaCl_2$

MC2 · 05.07.24 · $CaCl_2$

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 8.00

B = 0.51 µm/s

σ = 0.50

$WSSE_3$ch

97,213

Identifiability: OKAY

Channel side-split; high retentate-conc WSSE. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

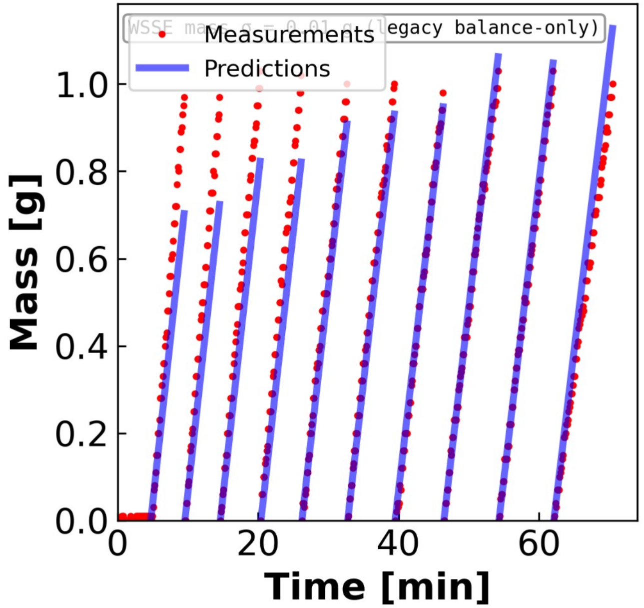
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

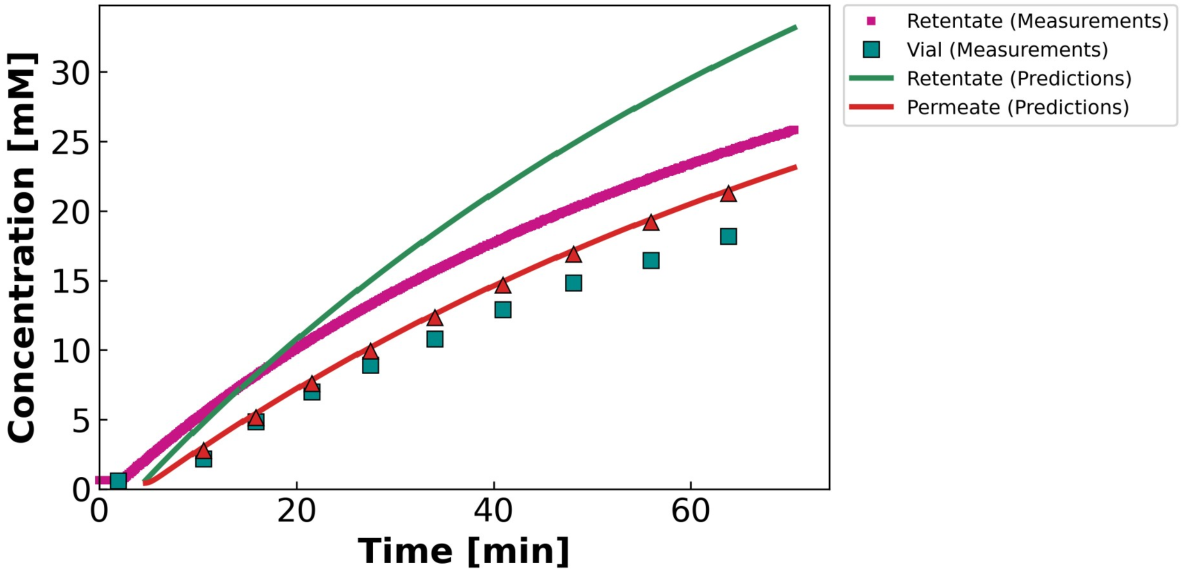
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 2 · $CaCl_2$

MC3 · 07.11.24 · $SCaCl_2$

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 6.56

B = 9.11 μm/s

$\sigma = 1.00$ ↿

WSSE$_3$ch

6,139

Identifiability: OKAY

σ at wall. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

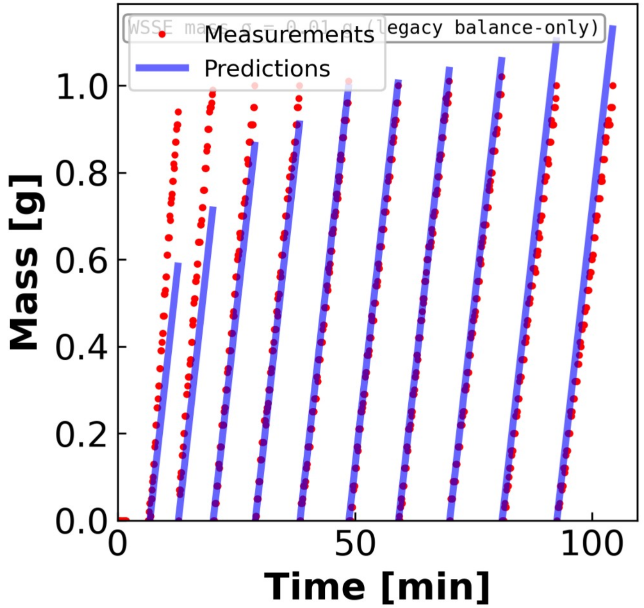
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

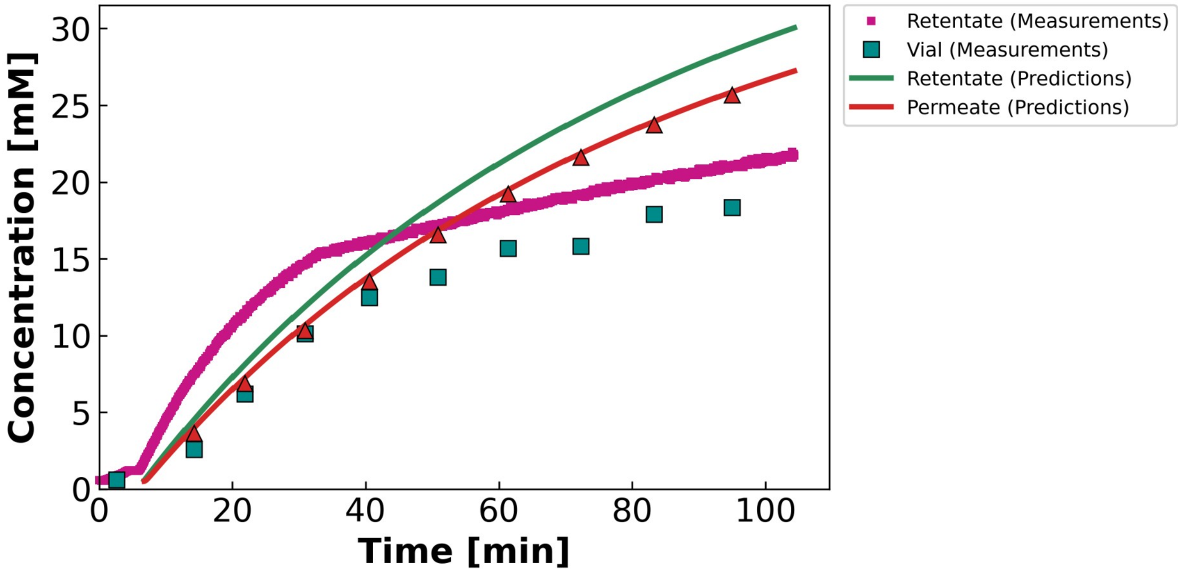
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 2 · $CaCl_2$

MC3 · 07.12.24 · S2$CaCl_2$

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 5.49

B = 30.00 ↑ µm/s

σ = 1.00 ↑

WSSE$_3$ch

9,774

Identifiability: OKAY

σ AND B pinned at walls (B=30). Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

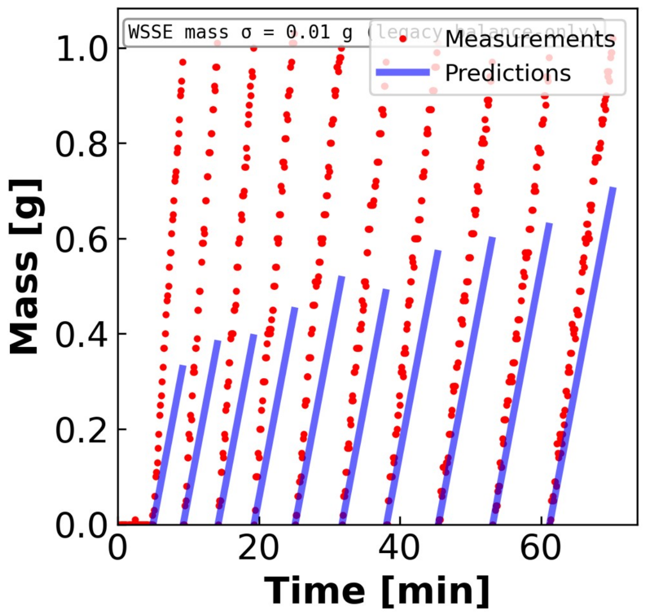
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

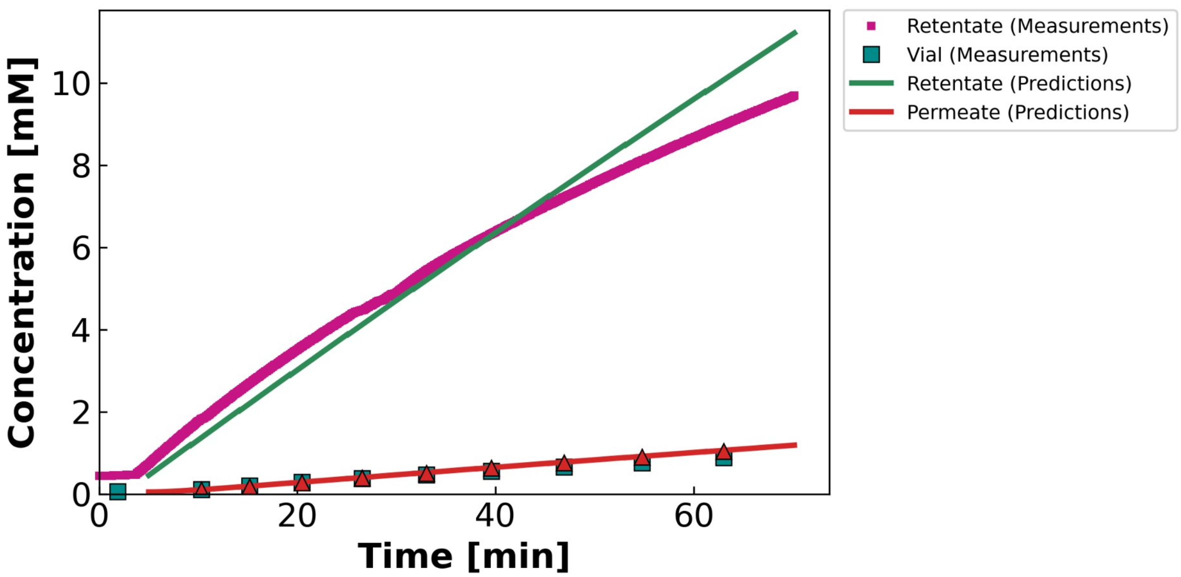
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 3 · $LaCl_3$

MC2 · 05.21.24 · $LaCl_3$

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 2.95

B = 0.31 μm/s

σ = 0.00 ↓

$WSSE_3$ch

21,500

Identifiability: POOR

Wrong-model bias — model can't track c^f. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

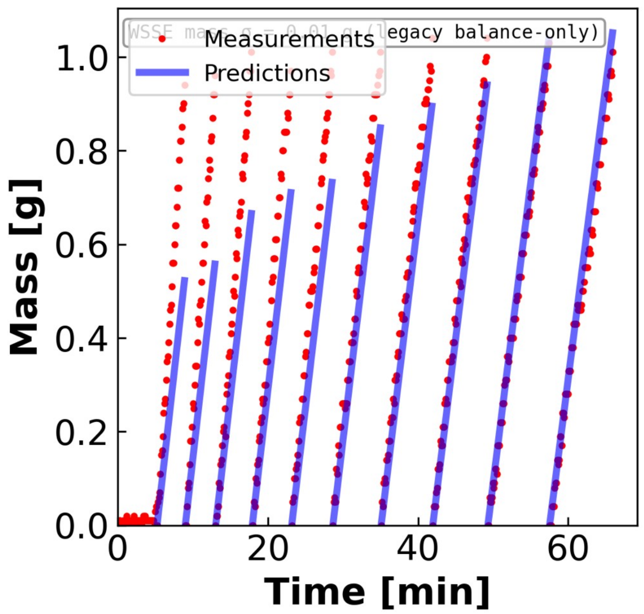
Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

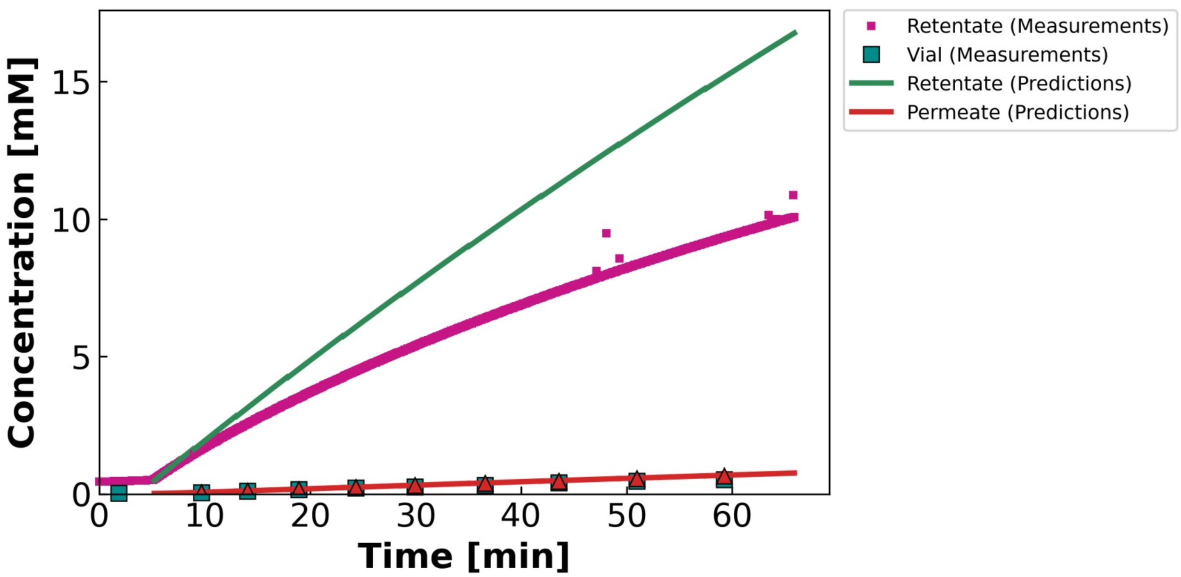
Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

STAGE 3 · $LaCl_3$

MC4 · 07.11.24 · $SLaCl_3$

Mass & concentration per vial · DATA2-style · prediction shown only where permeate is collected





θ (contour-consistent)

Lp = 5.55

B = 0.17 µm/s

σ = 0.93

WSSE_3ch

75,080

Identifiability: POOR

Replicate wrong-model bias. Variable B (B_form = 1): the concentration curve bends — a constant-B model would draw a straight line here.

Leading hold-up / startup vial excluded (n_{v0} = 2): the model integrates J_w across the hold-up to carry the cell state, but mass is plotted only where permeate is collected in the vial — DATA1/DATA2 convention.

Concentration legend is built into the panel (retentate / vial / permeate, measurement vs prediction).

05 • THE TAKEAWAY

Combine what we have, then design what we're missing

Q1 — a better initial guess from the existing runs · Q2 — the experiments that close the gaps

Take each parameter from the run that pins it

Same θ = (L_p, B, σ), but each component seeded from the experiment whose data actually constrain it

L_p

$J_w = L_p \Delta P$

from: Pure-water run, or any diluting run's mass channel

$\Delta\pi \approx 0 \rightarrow L_p$ falls straight out of the mass data, untouched by σ or B.

σ

argmin_σ WSSE

from: A high-cf (diluting) run — MC3 SNaCl, σ ≈ 0.79

Only where $\Delta\pi/\Delta P$ is appreciable does σ leave an interior minimum.

B

$J_s = B\,(c_{in} - c_h)$

from: The low-residual solute channel where Jw spans a range

Tighten with the per-salt bound (halve per +1 cation charge).

θ_init per salt = (L_p : water / diluting · σ : σ-probing run · B : clean solute channel)

Transfer across salts of the same valence, then run the joint multistart from this seed instead of a blind LHS.

New experiments, chosen to fix exactly what the math says is missing

Each one targets a specific identifiability gap from the sensitivity analysis

PRIORITY 1

→ pins L_p

Pure-water L_p before & after every run

$$Jw = Lp\ \Delta P$$

$\Delta\pi = 0$ isolates L_p so σ and B can't compensate for it. Doubles as a fouling check (pre/post drift).

PRIORITY 2

→ separates σ–B

Multi-ΔP sweep within a run + dilute runs

$$\partial Jw/\partial\sigma = -Lp\ \Delta\pi$$

Stepping ΔP changes Jw, shifting how σ (via Jw) and B (via Js) each act on c_f — breaking their correlation. Dilute runs raise Δπ/ΔP so σ identifies for $CaCl_2$/$LaCl_3$.

PRIORITY 3

→ tests model form

More $LaCl_3$ + retentate conductivity + ICP speciation

$$n\_eff = ?\ \ (La^{3+} \rightleftharpoons LaCl^{2+})$$

n = 2 is an anecdote (need ≥5). Retentate-side conductivity kills the $CaCl_2$ side-split; speciation tests the ion-pairing hypothesis directly.

QUESTIONS FOR DISCUSSION

What I'd like your read on before the re-run

1. Does the 3-stage diagnosis match your physical intuition? Increasing cation valence → identifiability degrades. Is the $\sigma=1$ wall a regime problem, or the c^f weighting forcing it there?

2. Two paths to well-identified — MC3 SNaCl (interior $\sigma = 0.79$) vs MC5 S2NaCl (diluting, σ irrelevant). What in the run-sheet metadata differs from the other NaCl sheets?

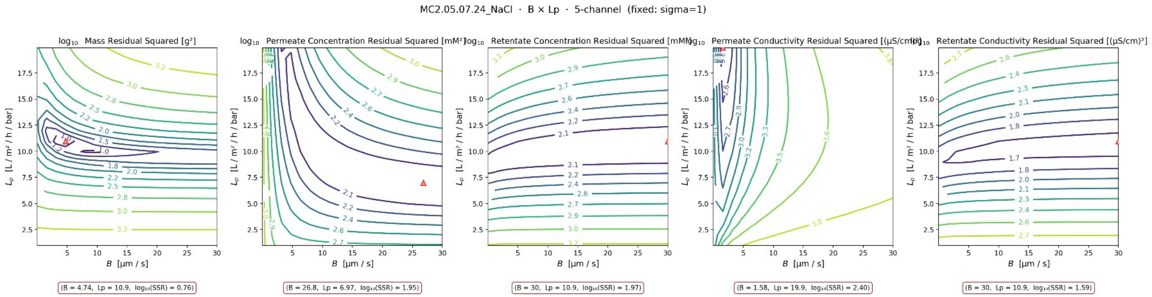
3. MC4 SNaCl is diluting but fits ~10× worse (WSSE ≈ 19.5k). Likely a data-quality issue (fouling / leak / calibration) rather than model — worth a closer look?

4. $LaCl_3$ fix: parameterize $\sigma(c^f)$, add an ion-pairing equilibrium block, or fit only the diluting portion? Either way we need 3–5 more sheets.

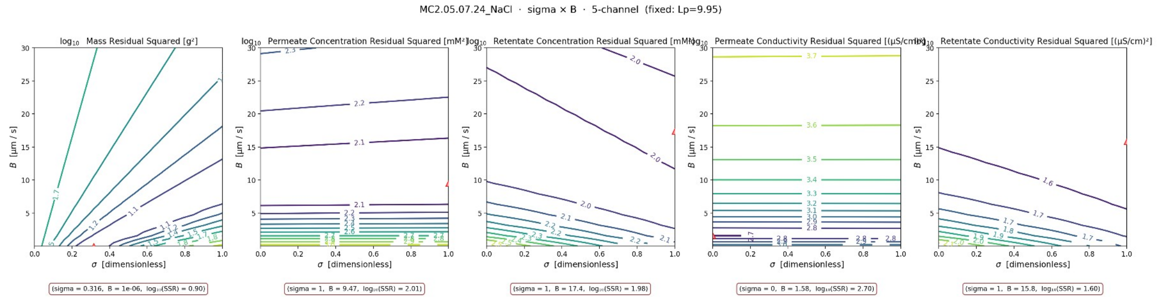
5. Per-salt B bounds — NaCl [0,30], $CaCl_2$ [0,15], $LaCl_3$ [0,10]. Anything you'd revise before re-running the full campaign?

MC2 NaCl — B-direction contours behind the σ=1 wall

The σ=1 wall on the Stage-2 NaCl sheets hides a B–L_p trade-off



$B \times L_p$ (σ fixed at warm-start = 1) — L_p on Y



σ × B (L_p fixed) — B on Y per axis convention